

Executing the Grey List: Interpretable Queries for Unfair-Term Detection in Terms of Service

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Abstract. The Annex of Directive 93/13/EEC, known as the grey list, describes the kinds of contractual terms that may be regarded as unfair. Automated detectors of unfair terms in online terms of service do not use it: they classify sentences with learned models whose decisions cite no legal ground. We ask what happens when the grey list is executed literally. Each clause of a contract is described by a simple template recording who may do what, under which conditions, with what notice and with what remedy for the other party. Templates are extracted by a language model that sees neither the queries nor the reference labels. Grey-list items are then written as queries over these templates and frozen before any evaluation, so that the result measures the list itself rather than our ability to fit it. On held-out contracts from the CLAUDETTE corpus, most items can be expressed this way, and the queries are precise on several of them. They retrieve, however, only a minority of the annotated unfair terms, and their overall performance stays below that of the clause theme alone and well below that of a fine-tuned legal language model. In exchange, every flag cites the grey-list item, the template fields and the sentences that triggered it, and the error analysis shows where a literal reading of the Annex and the benchmark diverge. Queries, templates and code are released.

Keywords. unfair contract terms, terms of service, Directive 93/13, legal knowledge representation, explainable AI, information extraction

1. Introduction

Directive 93/13/EEC on unfair terms in consumer contracts [1] defines unfairness by a general test—a significant imbalance in the parties’ rights and obligations, contrary to good faith—and complements it with an Annex: an indicative and non-exhaustive list of seventeen kinds of terms, item (a) to item (q), that *may* be regarded as unfair. Practitioners call it the grey list. It is the closest thing European consumer law offers to an operational definition of the phenomenon, and it is written in terms of *who* may do *what* to *whom* under *which* conditions: a seller who may terminate a contract of indeterminate duration

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without reasonable notice (*g*), alter the terms unilaterally without a valid reason specified in the contract (*j*), or exclude liability for personal injury (*a*).

Automated detectors of unfair terms do not use it. Since CLAUDETTE [2], the reference benchmark for online terms of service (ToS), the task has been framed as sentence classification: a learned model assigns one of eight unfairness categories to each sentence, and its decision cites no legal ground. Rationale-based extensions [3,4] attach explanations to predictions, but the explanations are learned too. Large language models prompted on the same task are neither more accurate nor more accountable [5,6].

This paper asks a simple question: **what happens if the grey list is executed literally?** We turn each item into a query over a structured representation of the contract: a *clause template* recording who acts, what is obliged, permitted or forbidden, and under which conditions, deadlines and remedies, attached to clauses identified by a human-validated thematic layer over the 50 CLAUDETTE contracts. That layer is the resource built and validated in our companion short paper [7], which asks how reliably legal themes can be annotated; the present paper is its sequel and asks what the themes make possible, namely the execution of the law itself. Queries are written before any template is extracted and frozen before any computation on held-out contracts, so that the result is a measurement of the grey list and not of our ability to fit it. Three questions structure the paper, each with a hypothesis recorded in advance (Table 3). **RQ-A** (Section 6): which items can be expressed, and do the frozen queries retrieve the reference labels better than the clause theme alone? **RQ-B** (Section 7): what does this interpretability cost against learned text-only models? **RQ-C** (Section 8): what do the false positives reveal about the benchmark?

Our contributions are: (i) an executable, frozen encoding of the grey list, with an analysis of which items can be expressed as template conditions; (ii) a pre-registered, per-item evaluation against reference labels on held-out contracts, including the gain over the clause theme alone; (iii) a cost accounting of interpretability against text-only models trained on the same split; (iv) an error analysis in which every flag is traceable to the fields and sentences that triggered it; and (v) the release of queries, hashes, templates and code.

2. Background and Related Work

Unfair-term detection. CLAUDETTE [2,8] labels sentences of 50 ToS with eight categories—arbitration, unilateral change, content removal, jurisdiction, choice of law, limitation of liability, unilateral termination and contract by using—whose definitions are themselves derived from the Annex and from doctrine [9]. The corpus has been extended to several languages [10,11], included in LexGLUE [12], and used to compare fine-tuned encoders with prompted LLMs [5]. Related resources annotate clause validity in German consumer contracts [13] and potentially abusive clauses in Chilean ToS [14]. All of them predict; none of them cites the legal ground of a prediction.

Legal knowledge representation. Rule languages such as LegalRuleML [15] and contract specification languages such as Symboleo [16] formalise norms as deontic statements with parties, actions and conditions; they are designed for authoring and reasoning, not for being populated from natural-language contracts at scale. Deontic modality extraction [17,18] recovers obligations, permissions and prohibitions from text, with

macro-F1 in the 0.6–0.8 range depending on the setting—good enough to be an input signal, not good enough to be a ground truth. Large language models annotate legal text reliably enough in zero-shot settings to serve as extractors [19]; we use one under a closed schema, as an extractor, never as a judge of unfairness. Our templates sit between these two traditions: a fixed, shallow structure that lawyers can validate clause by clause and that queries can be written against.

Queries over contracts. ContractNLI [20] evaluates document-level hypotheses (“the receiving party may share confidential information with employees”) by learned entailment; CUAD [21] retrieves clause types for due diligence. Both ask whether a contract contains a provision of a given kind, but answer with a learned model; we answer by executing an explicit condition and returning the template fields that satisfy it.

Explanation in AI and law. The field has long argued that legal decision support must explain in legal terms [22]. For unfair-term detection, explanations have so far been rationales selected by attention [3] or generated by a model [4]. A query that fires because a termination clause grants the provider a discretionary right without notice explains itself by construction.

3. The Executable Grey List: from the Annex to Frozen Queries

Table 1 lists the seventeen items of the Annex, the CLAUDETTE category each item most directly corresponds to, the clause theme under which such a term is found, and whether the item can be expressed as a condition over the clause templates defined below. Three groups emerge. *Structural items* describe a configuration of actor, modality, action and condition that a template can carry directly: items (a), (b), (d), (f), (g), (j), (k), (l), (m), (p) and (q). *Quantitative items* require a judgement of proportion or reasonableness that no template field encodes—“disproportionately high compensation” (e), “unreasonably early” (h)—and can be expressed only with an explicit, declared threshold. *Procedural items* depend on facts outside the text, such as whether the consumer had a real opportunity of becoming acquainted with the terms (i), and can be approximated at best by a proxy (acceptance by mere use, the CLAUDETTE category “contract by using”). Several categories of CLAUDETTE have no counterpart in the Annex: choice of law is not listed, and jurisdiction clauses are covered only through the general hindrance of legal action in (q). Conversely, items (c), (n) and (o) describe terms that are rare in ToS and absent from the reference categories. The grey list and the benchmark therefore overlap without coinciding, and the evaluation must be read item by item.

Clause templates. A template is a flat record attached to a clause (a maximal run of sentences sharing a theme). Its fields are fixed and few: the *actor* (provider, user, third party); the *modality* (obligation, permission, prohibition, power); the *action*, drawn from a closed inventory per theme (terminate, suspend, modify terms, change price, exclude liability, cap liability, remove content, impose arbitration, choose forum, . . .); the *object* or affected party; the *conditions* under which the action may be taken (for cause, for a specified reason, at discretion, none stated); the *notice or deadline* (a duration, “reasonable”, none, or not stated); and the *remedy* left to the other party (refund, right to cancel, compensation, none, not stated). Two conventions matter for the queries. *Not stated* is a value, distinct from *none*: a termination clause that says nothing about notice is not the same as one that excludes it, and grey-list items are sensitive to the difference. And a

Table 1. The Annex of Directive 93/13/EEC (grey list), its correspondence with CLAUDETTE categories (A arbitration, CH unilateral change, CR content removal, J jurisdiction, LAW choice of law, LTD limitation of liability, TER unilateral termination, USE contract by using) and with clause themes, and whether the item is expressible as a template condition (S structural, Q quantitative with declared threshold, P procedural proxy, – not expressible from the text).

Item	Term that has the object or effect of...	CLAUDETTE	Clause theme	Expr.
(a)	excluding/limiting liability for death or personal injury	LTD	Limitation of liability	S
(b)	excluding/limiting the consumer’s legal rights in case of non-performance	LTD	Limitation of liability; Warranty	S
(c)	binding the consumer while the seller’s performance depends on his will alone	–	Preamble/scope	–
(d)	retaining sums paid when the consumer withdraws, without reciprocity	–	Fees and payment	S
(e)	requiring disproportionately high compensation from the consumer	–	Fees and payment	Q
(f)	dissolving the contract at the seller’s discretion without reciprocity	TER, CR	Termination; User content	S
(g)	terminating a contract of indeterminate duration without reasonable notice	TER	Termination	S
(h)	automatically extending a fixed-term contract with an unreasonably early deadline	–	Fees and payment	Q
(i)	binding the consumer to terms he had no real opportunity to know	USE	Preamble/scope	P
(j)	altering the terms unilaterally without a valid reason specified in the contract	CH	Modification of terms	S
(k)	altering unilaterally the characteristics of the product or service	CH	Modification of terms	S
(l)	determining the price at delivery or increasing it without a right to cancel	CH	Fees and payment	S
(m)	letting the seller decide conformity or interpret any term exclusively	–	Preamble/scope; Warranty	S
(n)	limiting the seller’s obligation to respect commitments of his agents	–	Third-party services	–
(o)	obliging the consumer to perform while the seller does not	–	–	–
(p)	transferring the seller’s rights and obligations without the consumer’s agreement	–	Preamble/scope	S
(q)	excluding or hindering legal action, imposing arbitration, restricting evidence	A, J	Arbitration and disputes	S

clause may carry several templates when it states several norms; queries are evaluated per template and aggregated per clause.

From an item to a query: item (g). The Annex targets terms “enabling the seller or supplier to terminate a contract of indeterminate duration without reasonable notice except where there are serious grounds for doing so”. Its constitutive elements map onto template fields: the provider acts (actor *provider*, modality *permission* or *power*, action *terminate* or *suspend*), the termination is tied to no ground (conditions $\in \{\text{at discretion, none stated}\}$) and no reasonable notice is given (notice $\in \{\text{none, not stated}\}$). The exception in the item itself (“serious grounds”) becomes a blocking condition: a template

whose condition is *for cause*, or that is stated as an exception to such a template, does not fire. Executing $Q(g)$ returns, for each hit, the template, the item and the sentences cited; Section 8 shows one.

Queries. In general, a query is a conjunction of conditions over template fields, optionally over several templates of the same document, that returns the templates satisfying it together with the sentences they were extracted from. The queries were drafted with a generative AI assistant from the text of the Annex and the CLAUDETTE category definitions, then reviewed and approved by the authors, before any template was extracted. They were revised once, on development contracts only: six action codes were added to the inventories, and two queries were extended, $Q(q)$ to contractual limitation periods and $Q(i)$ to acceptance deemed by continued use. The set was then frozen with a recorded hash, before any held-out template was extracted, and never edited afterwards. Three further examples, in prose:

- $Q(j)$ — theme *modification of terms*; actor *provider*; action *modify terms*; conditions $\in \{\text{at discretion, none stated}\}$ and no valid reason specified; a variant $Q(j')$ additionally requires remedy $\in \{\text{none, not stated}\}$, that is, no right to cancel.
- $Q(a/b)$ — theme *limitation of liability*; action *exclude liability* or *cap liability*; object includes personal injury or gross negligence, or the exclusion is unbounded (no carve-out for mandatory law).
- $Q(q)$ — an *obligation* or *prohibition* on the user to arbitrate, waive class action or jury trial, restrict evidence, bear the burden of proof or limit the claim period; or the provider’s *power* to impose arbitration or choose the forum.

Queries come in three families: *presence* of a forbidden configuration (the four above); *absence* of an element the theme’s schema expects (a modification clause with no notification mechanism and no notice); and *composition* across clauses of the same document (a discretionary termination right of the provider with no symmetric termination right for the user). All are scored at the sentence level through the sentences their templates cite.

4. Materials: Corpus, Thematic Layer and Clause Templates

Corpus. CLAUDETTE comprises 50 online terms of service split into 9,414 sentences, of which 1,032 (11.0 %) carry at least one of 1,137 unfairness labels. The 17 held-out contracts contain 4,078 sentences, 440 of them labelled unfair.

Thematic layer (companion short paper). The clause themes come from the resource built in our companion short paper [7], which we reuse here unchanged. Every one of the 9,414 sentences carries one primary theme and zero or more secondary themes from a closed vocabulary of 20 legal themes: the four themes where unfairness concentrates (governing law, modification of terms, termination, limitation of liability); the contractual machinery (preamble and scope, eligibility and account, fees and payment, warranty disclaimer, arbitration and disputes, third-party services, privacy and data, DMCA notices); the content and conduct themes (licence and IP, user content, acceptable use, feedback, promotions); and the framing themes (communications, meta, miscellaneous boilerplate). Three trained annotators labelled each sentence independently from a written codebook; four language-model judges labelled the same sentences but were kept

Table 2. Unfairness by clause theme on the 50 CLAUDETTE contracts (consensus theme; n sentences, share carrying a CLAUDETTE label, lift over the 11.0 % base rate). Themes with fewer than 20 sentences are omitted.

Theme	n	unfair	lift	Theme	n	unfair	lift
Governing law	164	54.3 %	4.95	Fees and payment	1,077	5.5 %	0.50
Modification of terms	392	49.0 %	4.47	Warranty disclaimer	613	5.4 %	0.49
Termination	422	43.4 %	3.96	Eligibility and account	761	4.2 %	0.38
Limitation of liability	604	37.2 %	3.40	Privacy and data	278	4.0 %	0.36
Arbitration and disputes	847	10.5 %	0.96	Licence and IP	1,225	3.4 %	0.31
User content	658	9.7 %	0.89	Acceptable use	911	3.4 %	0.31
Preamble and scope	677	9.5 %	0.86	Miscellaneous boilerplate	586	0.9 %	0.08
Third-party services	464	6.5 %	0.59	Meta, communications	184	0.0 %	0.00

out of the reference, which is resolved strictly between annotators by a cascade (strict agreement 46.8 %, majority 48.3 %, human arbitration 4.9 %), with individual votes preserved [23,24,25,26]. Agreement is α -MASI 0.658 on the 20 themes and 0.725 on their 11-theme consolidation, whose merges follow observed confusions but never cross unfairness strata; the leave-one-annotator-out human ceiling is κ 0.859. The short paper measures the reliability of this layer; the present paper uses it three times: the consensus primary theme delimits clauses (maximal runs of sentences sharing a theme) and selects the action inventory the extractor sees; each query is scoped to the theme under which its item is found (Table 1); and the theme alone is the first baseline that any structured method must beat. Themes are not unfairness labels, but they are far from independent of them: Table 2 gives, for each theme, the share of sentences carrying a CLAUDETTE label. The four high-stratum themes carry lifts between 3.4 and 5.0 over the 11.0 % base rate; the framing themes are almost never unfair.

Template extraction. Templates are extracted automatically; clauses longer than eight sentences are split into windows of at most eight. A language model (Claude Opus 5) receives the clause sentences, the clause theme, the closed action inventory for that theme and the output schema, and returns zero or more templates, each citing the sentences that support it. It runs in an isolated context that contains neither the queries nor any reference label. Outputs are checked for schema conformance and structural validity (cited sentences exist, actions belong to the inventory); a lexical anchoring check then re-sets to *not stated* any condition, notice or remedy value that no cue in the cited sentences supports. Templates were not reviewed by legal experts in this study (Section 9). On a development pilot of 100 clauses extracted three times, 98 % of outputs conformed to the schema, the decisive fields (actor, modality, action, condition, notice, remedy) agreed across runs with a mean Jaccard index of 0.81, and the sentences flagged by the frozen queries agreed across runs with Fleiss’ $\kappa = 0.86$. The held-out contracts were extracted twice. The pre-registered analysis uses the first pass: 1,087 clause windows, all schema-conformant, 99.9 % structurally and semantically valid, yielding 2,809 templates (2.58 per clause), with 9.7 % of windows carrying none. The second pass, evaluated with the same frozen analysis, serves as a sensitivity check: its decisive fields agree with those of the first with a mean Jaccard index of 0.77, and the sentences flagged by the queries agree with Fleiss’ $\kappa = 0.82$.

Table 3. Pre-registered questions, hypotheses and tests, recorded before any computation on the held-out contracts. Components requiring expert judgement are deferred to the follow-up study.

RQ	Hypothesis H1	Metric	Test
A	precision ≥ 0.60 on the mapped categories; recall ≥ 0.50 on LTD, TER and CH; F_1 gain over theme only on these three	per item and query: P, R, F_1	document-level bootstrap CI
B	loss in $F_1 \leq 0.05$ against the best text-only model (explanation sufficiency: deferred)	F_1 difference	non-inferiority, one-sided CI
C	$\geq 20\%$ of false positives are grounded flags missing from the reference (expert judgement: deferred)	false positives by item, theme, severity	descriptive only

5. Pre-registered Evaluation Protocol

Split. The 50 contracts are split by document into 33 development contracts, used to build the field inventories, to debug the extraction and to revise the queries once, and 17 held-out contracts on which all numbers are reported. **Pre-registration.** The template schema and the queries were frozen, with a recorded hash and date, before the held-out templates were extracted; the extraction prompt is versioned and hashed as well. One change to the originally planned protocol was recorded at the same time: templates are used as extracted, without expert review, so the pipeline is evaluated end to end. **Reference.** A sentence is positive for a CLAUDETTE category if it carries that label in the original corpus; a query is scored against the category or categories its grey-list item maps to (Table 1), and, for items without a CLAUDETTE counterpart, against any unfairness label. **Metrics.** A flagged template marks only the sentences it cites. Per query: precision, recall and F_1 at the sentence level; per item: the same, aggregating the item’s queries; overall: micro- and macro-averaged over items, with document-level bootstrap intervals; and, for comparison with learned models, a binary *any unfairness* F_1 . **Baselines.** (i) *Theme only*: the best achievable retrieval from the clause theme alone, $P(\text{unfair} \mid \text{theme})$ estimated leave-one-document-out; (ii) a lexical floor, TF-IDF with logistic regression; (iii) a fine-tuned legal encoder [27]. Both learned models are trained on the development contracts and scored on the same held-out sentences. **Error analysis.** False positives and false negatives are compared with the reference labels and categorised by item, theme and CLAUDETTE severity, without expert adjudication.

6. Results A: Expressibility and Per-Item Retrieval

The theme alone is a strong baseline. On the 50 contracts, a predictor that knows only the clause theme reaches an average precision (area under the precision–recall curve) of 0.404 against a base rate of 0.110, that is, 3.7 times chance; Table 5 reports instead its precision at the decision threshold on the held-out contracts.

Expressibility. Eleven of the seventeen items are structural and expressible without a threshold; two require a declared threshold; one is a procedural proxy; three cannot be expressed from the text of the clause (Table 1). The items that carry most of the CLAUDETTE labels—(a)/(b), (f)/(g), (j)/(k) and (q)—are all in the structural group.

Per-item retrieval. Table 4 reports, for each item scored on the held-out contracts, precision and recall against the mapped reference categories and the difference with the

Table 4. Held-out retrieval per grey-list item (17 contracts, 4,078 sentences). n_+ : sentences carrying a mapped CLAUDETTE label; P, R, F_1 at sentence level with document-level bootstrap 95 % CI; theme only: P(label | theme) estimated leave-one-document-out; ΔF_1 : queries minus theme only. Items without a CLAUDETTE counterpart (flags: (d) 21; (e) 0; (h) 0; (m) 39; (p) 17) are not scored. Last row: micro-average over the structural items, which excludes the procedural proxy (i); macro- F_1 in parentheses.

Item	Ref.	n_+	Flagged	P	R	F_1 [95 % CI]	Theme	ΔF_1 [95 % CI]
(a)	LTD	132	4	0.50	0.02	0.03 [0.00; 0.08]	0.44	-0.41 [-0.50; -0.29]
(b)	LTD	132	19	0.79	0.11	0.20 [0.08; 0.31]	0.44	-0.24 [-0.40; -0.08]
(f)	TER,CR	140	12	0.83	0.07	0.13 [0.03; 0.26]	0.34	-0.21 [-0.33; -0.08]
(g)	TER	102	40	0.68	0.26	0.38 [0.27; 0.49]	0.36	+0.02 [-0.10; 0.13]
(i)	USE	44	37	0.43	0.36	0.40 [0.31; 0.46]	0.18	+0.21 [0.10; 0.30]
(j)	CH	88	42	0.62	0.30	0.40 [0.32; 0.50]	0.40	+0.00 [-0.10; 0.12]
(k)	CH	88	22	0.50	0.12	0.20 [0.08; 0.34]	0.40	-0.20 [-0.28; -0.11]
(l)	CH	88	18	0.61	0.12	0.21 [0.13; 0.29]	0.40	-0.19 [-0.33; -0.05]
(q)	A,J	45	103	0.17	0.38	0.23 [0.15; 0.32]	0.18	+0.05 [-0.04; 0.14]
Structural (8)				0.46	0.15	0.22 (macro 0.22)		-0.15

theme-only baseline. The queries are precise and sparse. Precision reaches 0.61 to 0.83 on five of the nine scored items ((b), (f), (g), (j), (l)), but recall never exceeds 0.38, and it stays at 0.30 on unilateral change, 0.26 on termination and 0.11 on limitation of liability. Over the eight structural items, that is, excluding the procedural proxy (i), micro-averaged precision is 0.46 and recall 0.15. The pre-registered hypothesis for RQ-A is therefore rejected: the recall threshold of 0.50 is missed on all three target categories. Against the theme alone, the queries are significantly better on one item only, (i), by +0.21 [0.10; 0.30] (Holm-adjusted permutation $p = 0.009$); they are indistinguishable on (g), (j) and (q), and significantly worse on (a), (b), (f) and (k); item (l) is also below the theme by its unadjusted interval (Table 4), but the difference does not survive the Holm correction ($p = 0.014$, adjusted 0.056). The gain on (i) is not robust to extraction noise: on the second extraction pass it shrinks to +0.14 [-0.01; 0.28] and loses significance (Holm-adjusted $p = 0.15$), while every other verdict is unchanged.

7. Results B: The Price of Interpretability

Table 5 compares the union of the frozen queries with three text-only systems on the same held-out sentences. The queries reach a binary F_1 of 0.36 [0.32; 0.42], with a precision of 0.41 (Wilson interval [0.36; 0.46]). The gap is -0.34 [-0.40; -0.28] against the fine-tuned legal encoder, -0.17 [-0.21; -0.12] against the lexical model and -0.07 [-0.12; -0.02] against the theme alone. Non-inferiority at the pre-registered margin of 0.05 is rejected in all three comparisons, and again on the second extraction pass, where the union reaches an F_1 of 0.34 [0.28; 0.41]. Of the 440 unfair sentences, the queries retrieve 145, of which the encoder also finds 123; 22 are found by the queries alone and 79 by neither system (Section 9).

Table 5. The price of interpretability on the held-out sentences (any CLAUDETTE label, 440 positives among 4,078). All systems are scored by the same evaluation run; the learned models are trained on the 33 development contracts.

System	P	R	F_1 [95 % CI]	Flagged	Cites a legal ground
Theme only (leave-one-document-out)	0.42	0.46	0.44	489	no
TF-IDF + logistic regression	0.45	0.63	0.53 [0.49; 0.57]	614	no
Legal-BERT, fine-tuned	0.64	0.77	0.70 [0.65; 0.75]	527	no
Frozen queries (union)	0.41	0.33	0.36 [0.32; 0.42]	356	yes

8. Results C: When a Query Fires and the Benchmark Does Not

A flag reads as follows. In one held-out contract, the sentence “We may terminate your access to and use of the services, at our sole discretion, at any time and without notice to you” yields the template {actor: provider; modality: power; action: terminate; condition: at discretion; notice: none; remedy: not stated}; Q(*g*) fires, the flag cites item (*g*) and this sentence, and the reference labels the sentence as unilateral termination. Every flag returns the template fields that satisfied the query and the sentences they were extracted from, so each error can be traced to one of three sources: a mis-filled field, a query broader than its grey-list item, or a missing reference label for a clause that a literal reading of the Annex captures. In this study the attribution is automatic and partial. **Misses.** No missed sentence lacks a template altogether: every one lies in a clause for which templates were extracted. For unilateral change, most misses carry no template with the item’s action. For limitation of liability and termination, about half carry the right action but a field that blocks the query, most often an exclusion whose object names neither personal injury nor gross negligence (42 sentences for (*a*) and (*b*)), or a termination read as requiring cause (15 for (*g*)). **False positives.** Item (*q*) accounts for 86 of the 162 item-level false positives, all in dispute-resolution clauses that CLAUDETTE leaves unlabelled at the flagged sentence. Across scored items, 66 item-level false positives lie in a clause whose other sentences carry the mapped label, which points to the sentence-level granularity of the reference rather than to the query. Finally, the items without a CLAUDETTE counterpart fire 77 times: 39 flags for (*m*), none of them labelled, 17 for (*p*) and 21 for (*d*). Whether such flags are grounded is left to the expert study (Section 9).

9. Discussion and Limitations

Four points follow. First, the grey list is executable, and executing it literally is precise but narrow. On five of the nine scored items, at least six flags in ten are correct, but the configurations the Annex describes appear to cover a small part of what the CLAUDETTE annotators labelled: our queries retrieve a third of it, and a fine-tuned encoder more than twice as much. Second, the two approaches fail on different sentences, so a grounded detector is not redundant with a learned one even where its aggregate recall is lower. Of the 145 unfair sentences the queries retrieve, 22 are missed by the fine-tuned encoder, and most are not borderline misses: for 13 of them the encoder’s unfairness score is below 0.1 and for 17 below 0.25. The queries have a different error profile, not simply a worse one; a naive union of the two systems, computed post hoc, reaches an F_1 of 0.61, below the encoder alone, so how to combine them remains to be designed. Within that profile,

the clause theme carries more retrievable signal than the templates add, since the only significant per-item gain does not survive a second extraction pass; what the templates contribute is *grounds*, the ability to say which item fired and why. The error analysis shows where the price of these grounds is paid: fields that block a query and actions missing from the templates, not absent templates. Third, and this is a position rather than a measured result, the two systems answer different questions asked by different actors. A regulator auditing a platform’s terms, or a consumer association preparing a complaint, needs a small number of precise, citable flags, and a precision above 0.6 on five of nine items is, we argue, enough to build a case on. A platform screening its own contracts at scale needs recall, and can absorb the manual review of ungrounded flags. Judged against the first use, a queries-only F_1 of 0.36 with legal traceability is not a worse system than a 0.70 black box; it is a different tool for a different actor. Fourth, the mismatch between the Annex and the benchmark is itself a finding. Categories such as choice of law have no grey-list ground; items such as *(d)*, *(m)* and *(p)* fire on the held-out contracts but have no CLAUDETTE counterpart, so a detector that executed the law faithfully would be penalised by the benchmark whenever it is right. **Limitations and future work.** The templates were not reviewed by legal experts within the time frame of this study. As a consequence, low recall on a given item cannot be attributed with certainty to the query’s narrowness rather than to extraction noise on the relevant field; the two are conflated in this end-to-end evaluation and will be disentangled by the expert validation described below. Our results characterise the pipeline as a whole and do not establish the correctness of individual templates. The extractor matters: on the development pilot, with the previous prompt version, a second model (gpt-6-astra) agreed with the first on the decisive fields with a mean Jaccard index of about 0.6. The held-out stability estimate rests on a single pair of extraction runs. The reference labels are sentence-level while the Annex is clause-level; thresholds for quantitative items are declared, not derived; the held-out set is small and the intervals are wide; the queries reflect one reading of one legal system’s list. This work is part of an ongoing collaboration with legal experts: the legal co-authors helped design the architecture of the system and brought the legal practitioner’s perspective to it. The follow-up study addresses these limitations: independent double validation of the templates by two legal experts, with per-field agreement, which will separate extraction errors from query errors; re-extraction with pinned model versions and replication with an open-weight model; and an expert audit of the flagged sentences and of the explanations the queries return.

10. Conclusion and Availability

We have executed the grey list of Directive 93/13/EEC, a list of configurations of rights and obligations, as frozen queries over clause templates extracted automatically from the CLAUDETTE corpus. The result is a detector whose every decision cites the item it implements, and an item-by-item measurement of how much of the annotated unfairness a literal reading of the Annex retrieves: on held-out contracts, a precise but narrow fraction, about a third, below the clause theme alone and well below a learned encoder. With the companion short paper, which releases the thematic layer and its codebook, we release the queries with their hashes, the extracted templates, the code and the annotation platform.

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